Information Sciences 758 (2027) 123973

Contents lists available at ScienceDirect

Information Sciences

journal homepage: www.elsevier.com/locate/ins

Integrating cognitive skill levels assessment cognitive diagnosis for intelligent education systems

Xiaonan Yang, Senlin Luo, Jinjie Zhou, Deshan Yang, Limin Pan*

Information System and Security & Countermeasures Experimental Center, Beijing Institute of Technology, Beijing 100081, China

ARTICLE INFO

Keywords:
Cognitive diagnosis
Cognitive skills
Knowledge structures
Intelligent educational systems

ABSTRACT

Cognitive diagnosis enables assessing learners' cognitive states based on their performance. However, precisely modeling the relationship between learners' external performance and their underlying cognitive states, as well as inferring proficiency on unlearned concepts remains thorny task. Existing methods focus on assessing learners' knowledge proficiency, ignoring the impact of cognitive skill levels on performance, which yields biased fit of interaction process and suboptimal diagnostic accuracy. Moreover, current methods introduce prerequisite relations to infer proficiency on unlearned concepts, yet relational information weakens with increasing inference path length, leading to unreliable outcomes. This work proposes an Integrating Cognitive Skill Levels Assessment Cognitive Diagnosis (ICSCD). An exercise–cognitive level mapping is established to distinguish the cognitive skills examined by the exercises. Then a progressive interaction network is developed to differentially simulate learners' interaction process with exercises at different cognitive levels, alleviating the bias of interaction simulation. Furthermore, a node embedding representation method incorporating topic associations is devised, which constructs topic associations between learned concepts and unlearned concepts, reducing the loss of relational information, thereby obtaining reliable proficiency on unlearned concepts. Experimental results demonstrate ICSCD outperforms the state-of-the-art methods, enabling detailed identification of learners' cognitive strengths and weaknesses.

1. Introduction

Intelligent Tutoring Systems (ITS) [1,2] provide access to personalized learning services, which considerably advance the burgeoning growth of online education. Cognitive diagnosis [3–5] is a crucial component of ITS that aims to assess individuals' or groups' underlying cognitive states based on their performance, which provides insight into learners' cognitive deficits and serves as a basis for personalized instructions, such as recommendations for tailored learning resources and pathways [6,7].

A pivotal aspect of cognitive diagnosis lies in the design of appropriate interaction functions that model learners' cognitive process during problem-solving. This modeling aims to establish the correlation between learners' observable performance and their latent cognitive states, thereby enabling accurate diagnostic outcomes. Traditional cognitive diagnosis models, such as Item Response Theory (IRT) [8] and the Deterministic Input, Noise, and Gate (DINA) [9] model rely on manually crafted interaction functions to simulate learner-exercise interactions. However, manual functions exhibit limited fitting capabilities, struggling to capture the complex, nonlinear interactions between learners and exercises [10,11]. To alleviate this limitation, NCDM [10] pioneers the use of neural networks to learn interaction functions, significantly enhancing the capacity to model intricate cognitive process. GCDM [12]

* Corresponding author.
Email addresses: xinay2019@163.com (X. Yang), luosenlin2019@126.com (S. Luo), zhoujinjie@bit.edu.cn (J. Zhou), dsyang@canre.org.cn (D. Yang), panlimin@bit.edu.cn (L. Pan).

https://doi.org/10.1016/j.ins.2026.123973
Received 24 June 2025; Received in revised form 29 June 2026; Accepted 1 August 2026
Available online 3 August 2026
0020-0255/© 2026 Published by Elsevier Inc.

Table 1
An example of two exercises with different examination manners corresponding to different cognitive skills.

Exercise Content	Concept	Cognitive skills
A: Briefly explain what Newton's second law is?	Newton's second law	remembering
B: A car with a mass of 4 tons is traveling on a level road. If the car is subject to a traction force of 1000N, how much acceleration will it produce? (Suppose the resistance of the car is 0)	Newton's second law	applying

and RCD [13] further introduce graph neural networks to model the relational structure among learners, exercises, and knowledge concepts, thereby improving diagnostic performance.

The above studies typically rely on a binary exercise–concept mapping matrix and use multi-layer perceptron to construct an interaction function for assessing learners' proficiency in concepts. However, they lack the ability to assess learners' cognitive skill levels (e.g., remembering and applying levels). The cognitive skill levels of learners have a significant impact on their performance. For instance, as shown in Table 1, even if two exercises target the same concept, a learner's responses to them can differ markedly due to discrepancies in cognitive skills. Existing methods typically represent a learner's cognitive state using proficiency in concepts alone, without distinguishing the role of different levels of cognitive skills in the response process, thereby reducing the accuracy and interpretability of diagnostic results. Further analysis reveals two major limitations in using existing methods to directly assess learners' cognitive skill levels: (1) the binary exercise–concept mapping matrix cannot differentiate exercises that assess the same concept but target different cognitive skills; (2) exercises examining higher-order cognitive skills (such as analyzing, evaluating, and creating) have relatively few response samples—for example, such exercises account for only about 10% of the data in the MOOCRadar dataset [14]. Since multi-layer perceptron fully shares parameters across all samples, this scarcity of data leads to underfitting on these exercises, thereby reducing the accuracy of the assessment.

Another crucial issue is that the number of exercises learners answer is usually limited [11,15]. As a result, learners may lack practice on certain concepts (called unlearned concepts). The model's diagnosis results are unreliable due to the lack of corresponding response data. A widely adopted strategy to tackle this issue is to infer learners' proficiency on unlearned concepts by leveraging their relationships with those that have been learned. For example, RCD [13] and QCCDM [15] introduce the expert-annotated concept prerequisite relation graph to assess proficiency on unlearned concepts. However, in the absence of direct prerequisite relations, these models rely on multi-hop transmission to establish indirect associations between unlearned concepts and learned concepts. This tends to dilute the strength of relational constraints, resulting in severe deviations in diagnostic results.

In light of these challenges, a novel **I**ntegrating **C**ognitive **S**kill Level Assessment **C**ognitive **D**iagnosis (ICSCD) is proposed. Inspired by Bloom's cognitive taxonomy [16], an exercise–cognitive level mapping is established to distinguish the cognitive skills examined by the exercises. We introduce cognitive proficiency, which quantifies learners' mastery of cognitive skills at different levels. Then a progressive interaction network is constructed, which gradually allocates model parameters to accurately simulate the differentiated interaction process at different cognitive levels. Furthermore, we introduce topic associations between unlearned and learned concepts to establish direct and reliable inference paths, which facilitate accurate assessment of proficiency on unlearned concepts. Extensive experiments conducted on real-world datasets demonstrate the superiority of ICSCD over the state-of-the-art methods.

The main contributions can be summarized as follows:

- A novel **I**ntegrating **C**ognitive **S**kill Level Assessment **C**ognitive **D**iagnosis (ICSCD) is proposed, which defines cognitive proficiency to assess learners' mastery of different cognitive skills, enabling a granular identification of strengths and weaknesses of cognitive skills.
- A progressive interaction network is developed to differentially fit the interaction process between learners and exercises at different cognitive levels, thus reducing the fitting bias.
- A concept node embedding representation that incorporates topic associations is devised, which constructs topic association constraints between learned concepts and unlearned concepts, enabling accurate proficiency assessment of unlearned concepts.
- Experimental results and analysis on several benchmark datasets demonstrate that ICSCD achieves superior performance compared to baseline methods.

A toy example is first employed to elucidate the inputs and outputs of our model, as depicted in Fig. 1. The subsequent sections of this paper are structured as follows: Section 2 presents an overview of related works. Section 3 details the principle of ICSCD. Section 4 provides the results and analysis of experiments. Discussions are presented in Section 5. Concluding remarks and future research directions are provided in Section 6.

2. Related work

2.1. Cognitive diagnosis models

The objective of cognitive diagnosis is to perform an assessment of learners' cognitive states based on learner-exercise interactions [5]. Cognitive diagnosis can be categorized into two main categories: traditional statistical methods and neural network-based methods.

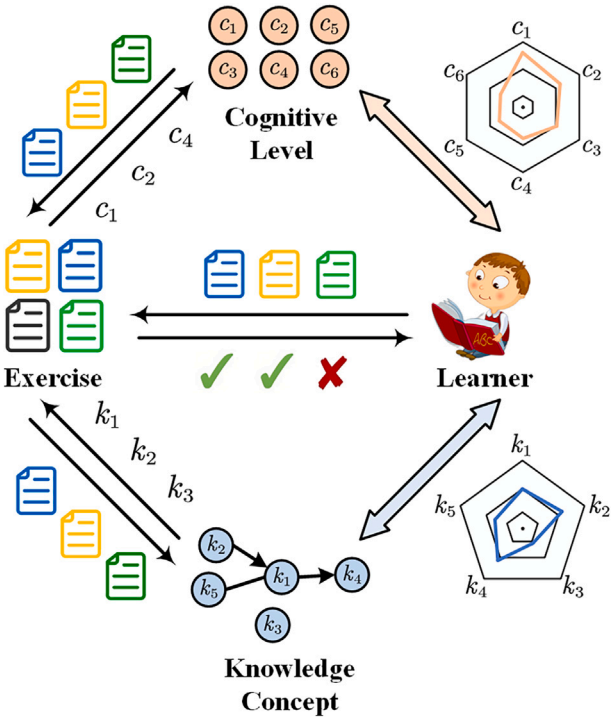


Fig. 1. A toy example of ICSCD, which exploits exercise records and attributes of exercises (the knowledge concepts and cognitive skills at different levels examined) to infer learners' cognitive states. Unlike existing works that only focus on the mastery of knowledge concepts (lower triangle), ICSCD also assesses the mastery of cognitive skills at different levels (upper triangle).

Traditional statistical methods such as IRT [8] and MIRT [17] aim to model the learner-exercise interactions via logistic-like functions. DINA [9] uses binary variables to diagnose the mastery of concepts considering learners' slip and guessing factors. [18] extended the DINA model to jointly fit item responses and response times, and Bayesian MCMC estimation verified that incorporating response time data can greatly improve classification accuracy of attribute patterns and the quality of parameter estimation.

Benefiting from the powerful modeling capabilities of deep learning [19], neural network-based methods have rapidly developed and achieved favorable results. A three-layer positive fully connected neural network is used by NCDM [10] to learn the complex interactions between learners and exercises. QRCDM [20] combines the explicit and implicit relationships between exercises and knowledge concepts, while quantifying the importance of concepts through an attention mechanism, thereby achieving higher prediction performance. CMNCD [21] explores the visual and textual content of rich cross-modal exercises and learns distinguishable exercise feature vectors to diagnose learners' knowledge proficiency. CMES [22] proposes a collaborative-aware mixed exercise sampling framework that can utilize the information of un-interacted exercises to enhance cognitive diagnosis tasks. ACD [23] introduces affect modeling into cognitive diagnosis by integrating students' affective states, enabling more accurate and interpretable diagnosis within a unified framework. ORCDF [24] utilizes learning response signals in the learning process through a response-aware graph convolution network, which effectively alleviates the over-smoothing problem and enhances discrimination on students' knowledge proficiency. PCDF [25] adopts the cumulative category response function theory to partition and merge data, enabling models to seamlessly analyze graded response data.

2.2. Bloom's taxonomy

Bloom's Taxonomy [16] describes a systematic framework that categorizes cognitive skills into six levels: remembering, understanding, applying, analyzing, evaluating, and creating, as shown in Fig. 2. These skills progress from simplicity to complexity, with mastery of each simpler skill serving as a prerequisite for mastering the subsequent more complex skill. In educational practice, exercises corresponding to different cognitive levels are designed to guide learners in mastering different cognitive skills. As exemplified in Table 1, Exercise A and B examine the same concept (Newton's second law), but the former focuses on examining learners' remembering skill, while the latter focuses on applying skill.

Learners' cognitive skill levels have a significant impact on their performance. Existing methods typically characterize a learner's cognitive state through their knowledge proficiency, without accounting for the critical role of cognitive skill levels during the problem-solving interaction process, thereby compromising the accuracy of diagnostic outcomes. Upon further analysis, two major challenges hinder the assessment of cognitive skills: (1) how to distinguish exercises that examine the same concept but different cognitive skills; (2) how to mitigate the model underfitting problem caused by the limited number of responses to exercises targeting higher-order cognitive skills (i.e., analysis, evaluation, and creation). This paper is dedicated to tackling these two issues.

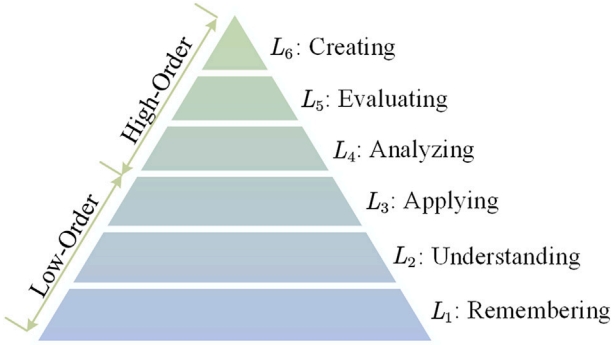


Fig. 2. Cognitive levels of Bloom's Taxonomy.

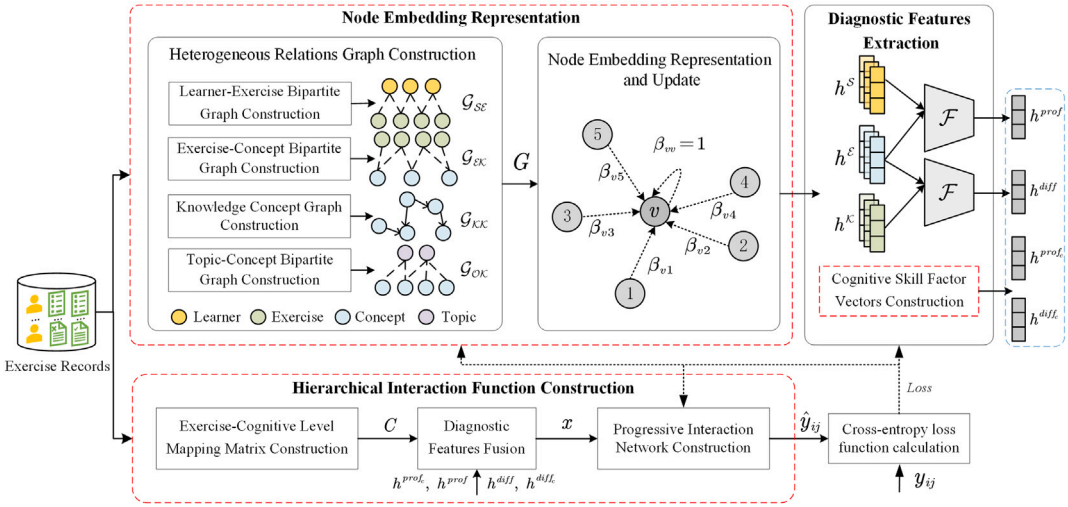


Fig. 3. The principal framework of the proposed ICSCD.

2.3. Knowledge structure

Knowledge structure typically manifests in the form of knowledge graphs, encompassing a plethora of domain-specific concepts and their relations. The profound exploration and application of knowledge structures are deemed crucial for assessing learners' mastery of concepts, with particular emphasis on the assessment of unlearned concepts [11]. For instance, RCD [13] models prerequisite relations between concepts by using multi-layer relation graphs, fostering diagnostic precision. CDGK [26] aggregates knowledge concepts into a graph structure by only utilizing leaf nodes of the knowledge tree to reduce dimensionality without losing accuracy, thereby achieving better accuracy. HCGCDM [27] leverages large language models to build a heterogeneous concept graph by extracting concept pairs and verifying relational triples, outperforming state-of-the-art models on multiple evaluation tasks. QCCDM [15] models concept causal relations by the structural causal model to infer learners' knowledge proficiency, thereby facilitating the assessment of unlearned concepts.

However, the lack of direct structural relations among many concepts necessitates multiple transmissions through intermediate concepts to establish relational connections. This process leads to the weakening of relational information, resulting in unreliable diagnostic results. Consequently, this paper establishes an association path between concepts under the same topic to enhance the correlation between unlearned and learned concepts and improve the accuracy of the assessment of proficiency in unlearned concepts.

3. Principle of ICSCD

The overall framework of ICSCD, as depicted in Fig. 3, is structured into three core modules: node embedding representation, diagnostic feature extraction and hierarchical interaction function construction. Specifically, in the node embedding representation module, a heterogeneous relations graph is first constructed. Then the heterogeneous graph attention network (HGAT) is leveraged to learn the node embedding representations and dynamically update them. In the diagnostic feature extraction module, learner's knowledge proficiency, cognitive proficiency, knowledge difficulty and cognitive difficulty are constructed to quantify the learner's cognitive ability and the difficulty posed by the exercises.

In the hierarchical interaction function construction module, an exercise-cognitive level mapping matrix is constructed to distinguish the cognitive skills examined by the exercises. Then the diagnostic features are fused to preliminarily simulate the interaction

process between learners and exercises. On this basis, a progressive interaction network is designed to adaptively allocate different sub-networks for interaction modeling according to the cognitive level to which the exercise belongs, thereby producing improved prediction performance. The cross-entropy loss function is used for model training to optimize cognitive proficiency and knowledge proficiency, and the final diagnosis results are obtained after the training is completed.

3.1. Preliminaries

Cognitive Diagnosis: An overview of the cognitive diagnosis task is first given. Suppose that the sets of learners, exercises, and concepts are respectively denoted as $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$, $\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$ and $\mathcal{K} = \{k_1, k_2, \ldots, k_K\}$, respectively. N, M,K are the sizes of each set. In accordance with the unique requirements of learner i, exercise j is selected to match their needs, resulting in corresponding answer y_{ij}. The exercise records are articulated as a set of triples $T = \{(s_i, e_j, y_{ij}) \mid s_i \in \mathcal{S}, e_j \in \mathcal{E}, y_{ij} \in \{0,1\}\}$, where $y_{ij} = 1$ means the answer is correct and $y_{ij} = 0$ means wrong. The relations between exercises and concepts are expressed as a binary matrix $Q = (Q_{jk} \in \{0,1\})^{M \times K}$ (annotated by experts), where the element Q_{jk} indicates whether exercise j contains concept k or not.

Knowledge Structure: The knowledge structure is evidenced by a knowledge concepts graph $\mathcal{G}_{\mathcal{KK}} = (\mathcal{K}, \mathcal{R}_{\mathcal{KK}})$, consisting of concepts $\mathcal{K}$ and the relations between them $R_{\mathcal{KK}}$.

Cognitive Level: Following Bloom's cognitive taxonomy [16], cognitive skills are delineated into six cognitive levels ranging from simple to complex. The mapping relations between exercises and cognitive levels can be denoted as $C = (C_{jl} \in \{0,1\})^{M \times 6}$. $C_{jl} = 1$ means that the highest level of cognitive skill examined by exercise j is l, otherwise $C_{jl} = 0$.

Knowledge proficiency and cognitive proficiency: Knowledge proficiency or cognitive proficiency refers to the learner's mastery level of a concept or cognitive skill. The value range is 0 to 1. The larger the proficiency value, the higher the mastery level.

Problem definition: Given learners' exercise records denoted as triplet logs T, a relation matrix Q between exercises and concepts, a relation matrix C between exercises and cognitive skill levels, and a knowledge structure represented by a knowledge graph $\mathcal{G}_{\mathcal{KK}}$, the goal is to diagnose the cognitive states of learners (both the proficiency in concepts and cognitive skills).

3.2. Node embedding representation

Existing graph-based methods typically rely on the relationships between concepts, updating the embeddings of unlearned concept nodes by aggregating information from learned concepts to assess the learner's proficiency on unlearned concepts. However, since the strength of inter-concept associations decays as the path length increases, it becomes challenging for the model to effectively incorporate information from distant learned concepts. This limitation hampers the accurate representation of unlearned concept nodes and thus affects the reliability of proficiency assessments on unlearned concepts.

To address this issue, by establishing the association path between learned and unlearned concepts within the same topic, we leverage the semantic relatedness of concepts to enhance the associations between learned and unlearned concepts. A graph attention network is then employed to perform weighted aggregation of information from neighboring nodes, which enriches the feature representation of unlearned concepts, thereby improving the accuracy of learners' proficiency in unlearned concepts.

3.2.1. Heterogeneous relations graph construction

Based on the exercise–concept mapping matrix Q, an exercise–concept bipartite graph $\mathcal{G}_{\mathcal{EK}}$ is constructed, and a learner–exercise bipartite graph $\mathcal{G}_{\mathcal{SE}}$ is built using the exercise records T. Considering that concepts under the same topic often exhibit strong semantic relevance—for example, "factoring" and "completing the square" both belong to the topic "solving quadratic equations"—a topic–concept bipartite graph $\mathcal{G}_{\mathcal{OK}}$ is introduced to enhance the associations between concepts. Then the $\mathcal{G}_{\mathcal{SE}}$, $\mathcal{G}_{\mathcal{EK}}$, $\mathcal{G}_{\mathcal{KK}}$ and $\mathcal{G}_{\mathcal{OK}}$ are integrated into a unified heterogeneous graph $\mathcal{G}$ to comprehensively model the multi-level structural dependencies among learners, exercises, concepts, and topics. As depicted in Fig. 3, $\mathcal{G} = \{\mathcal{V}, \mathcal{R}\} = \{(v, v', r) \mid v, v' \in \mathcal{V}, r \in \mathcal{R}\}$, where $\mathcal{V}$ and $\mathcal{R}$ refer to the sets of nodes and edges. The set $\mathcal{V} = \mathcal{S} \cup \mathcal{E} \cup \mathcal{K} \cup \mathcal{O}$, contains four types of nodes: learner $\mathcal{S}$, exercise $\mathcal{E}$, concept $\mathcal{K}$ and topic $\mathcal{O}$, and $\mathcal{R} = \mathcal{R}_{\mathcal{SE}} \cup \mathcal{R}_{\mathcal{EK}} \cup \mathcal{R}_{\mathcal{KK}} \cup \mathcal{R}_{\mathcal{OK}}$ represents their relations.

3.2.2. Node embedding representation and update

The initial embedding representations for each type of node are computed as follows:

$$h_i^{\mathcal{S}} = x_i^{\mathcal{S}} \times U,\ h_i^{\mathcal{E}} = x_i^{\mathcal{E}} \times B,\ h_i^{\mathcal{K}} = x_i^{\mathcal{K}} \times D,\ h_i^{\mathcal{O}} = x_i^{\mathcal{O}} \times Z \tag{1}$$

where $h_i^{\mathcal{S}}$, $h_i^{\mathcal{E}}$, $h_i^{\mathcal{K}}$ and $h_i^{\mathcal{O}}$ denote the initial node embedding representations of the learner, exercise, concept and topic i, respectively; $x_i^{\mathcal{S}}$, $x_i^{\mathcal{E}}$, $x_i^{\mathcal{K}}$ and $x_i^{\mathcal{O}}$ represent the one-hot encoded vectors of the learner, exercise, concept and topic i, respectively; U, B, D, and Z are four trainable matrices.

Following the paradigm of the HGAT [28], a dual-level attention mechanism is used to capture the importance attributed to both the node level and the relation level. Considering a specific node $v \in \mathcal{V}$, the significance of various relation types is first acquired through the relation-level attention mechanism, which can be formulated as follows:

$$h_r = \sum_{v'} A_{vv'} h_{v'} \tag{2}$$

$$a_r = \sigma\left(\mu_r^{\top} \cdot \left[h_v, h_r\right]\right) \tag{3}$$

$$\alpha_r = \frac{\exp(a_r)}{\sum_{r' \in \mathcal{R}_v} \exp(a_{r'})} \tag{4}$$

where h_r denotes the embedding of the relation r; h_v, $h_{v'}$ represent the embedding of node v and the embedding of neighboring node v' that has the relation r with v; $A_{vv'}$ is the value of v-th row and v'-th column of A. A is the normalized adjacency matrix with self-connections, whose rows represent all nodes and columns represent their neighboring nodes. The relation-level attention score a_r is calculated with h_r and h_v, where $[\cdot]$ signifies concatenation, $\sigma(\cdot)$ represents the activation function and μ_r^T is the trainable attention vector for the relation r.

For different types of relations $r' \in \mathcal{R}_v$, where $\mathcal{R}_v$ represents the set of all relation types related to node v, the corresponding relation-level attention scores of different relations are calculated respectively. The softmax function is then applied to obtain the weighted relation-level attention score α_r.

Based on the relation-level attention, the node-level attention is employed to capture the importance of different neighboring nodes with the relation type r', which can be formulated as follows:

$$b_{vv'} = \sigma\left(\xi^T \cdot \alpha_{r'} \left[h_v, h_{v'}\right]\right) \tag{5}$$

$$\beta_{vv'} = \frac{\exp(b_{vv'})}{\sum_{v'' \in \mathcal{N}_v^{r'}} \exp(b_{vv''})} \tag{6}$$

where ξ is the trainable attention vector and $\alpha_{r'}$ stands for the weighted relation-level attention score of the relation r'; $b_{vv'}$ denotes the node-level attention score of node v and v'; $\mathcal{N}_v^{r'}$ is the set of neighboring nodes of node v with the relation r'. Similarly, the softmax function is used to normalize the node-level attention scores and finally obtain the weighted node-level attention scores $\beta_{vv'}$.

The relation-level and node-level attention mechanisms are integrated into node information aggregation, where weighted averaging is adopted to adaptively learn the importance of individual nodes and their relations:

$$h_v^{(d+1)} = \sigma\left(\frac{1}{|\mathcal{N}_v| + 1}\left(h_v^{(d)} + \sum_{v' \in \mathcal{N}_v} \beta_{vv'} \cdot h_{v'}^{(d)}\right)\right) \tag{7}$$

where d represents the index number of layers of heterogeneous graph attention networks and $|\mathcal{N}_v|$ is the size of the neighbor node set of node v. Finally, the learner, exercise, concept, and topic node embedding representations $\mathcal{H}^S$, $\mathcal{H}^E$, $\mathcal{H}^K$ and $\mathcal{H}^O$ are obtained.

3.3. Diagnostic features extraction

The concept node embeddings are combined with the learner and exercise embeddings, and are fed into the multi-layer perceptron (MLP) to extract h^{prof} and h^{diff}. $h^{prof} \in (0,1)^{N \times K}$ represents knowledge proficiency, and the i-th row represents the mastery of each knowledge concept by learner i. $h^{diff} \in (0,1)^{N \times K}$ represents knowledge difficulty, and the j-th row represents the difficulty of each concept examined by exercise j.

$$h^{prof} = \mathcal{F}\left(\left[h^S, h^{\mathcal{K}}\right]\right) \tag{8}$$

$$h^{diff} = \mathcal{F}\left(\left[h^{\mathcal{E}}, h^{\mathcal{K}}\right]\right) \tag{9}$$

The specific MLP, represented by $\mathcal{F}$, comprises a sequence of two linear layers accompanied by Sigmoid activation functions applied to each layer. Based on Bloom's taxonomy [16], we construct cognitive skills factor vectors: the learner's cognitive proficiency vector h^{prof_c} and the exercise's cognitive difficulty vector h^{diff_c}. For learner i and exercise j, the $h_i^{prof_c}$ and $h_j^{diff_c}$ are defined as follows:

$$h_i^{prof_c} = \sigma\left(\boldsymbol{x}_i^S \times P^{prof}\right), \quad h_j^{diff_c} = \sigma\left(\boldsymbol{x}_j^{\mathcal{E}} \times P^{diff}\right) \tag{10}$$

where $\boldsymbol{x}_i^S$ and $\boldsymbol{x}_j^{\mathcal{E}}$ are the one-hot encoded representations of learner i and exercise j, respectively; P^{prof} and P^{diff} are learnable matrices; σ is the Sigmoid activation function. $h_i^{prof_c} \in (0,1)^{1 \times 6}$ represents learner i's proficiency at six levels of cognitive skill, while $h_j^{diff_c} \in (0,1)^{1 \times 6}$ reflects the difficulty of exercise j with respect to six cognitive skills.

3.4. Hierarchical interaction function construction

Existing methods typically rely on MLPs to construct interaction functions that assess learners' knowledge proficiency, neglecting the assessment of learners' cognitive skill levels. This oversight leads to notable discrepancies between predicted and actual performance, ultimately degrading the final diagnostic accuracy.

To address this issue, we propose the following solutions: (1) Guided by Bloom's taxonomy, exercises are classified into six levels of cognitive skills, providing a labeled basis for assessing learners' cognitive skill levels; (2) Considering that solving exercises requiring higher-order cognitive skills involves lower-order skills as prerequisites, we utilize the assessment of lower-order cognitive skills to assist in the assessment of higher-order ones. The core contribution of this module is to construct an exercise–cognitive level mapping matrix to distinguish cognitive skills across exercises, and to design a progressive interaction network that assesses learners' cognitive skill levels from lower to higher levels, thereby improving diagnostic accuracy.

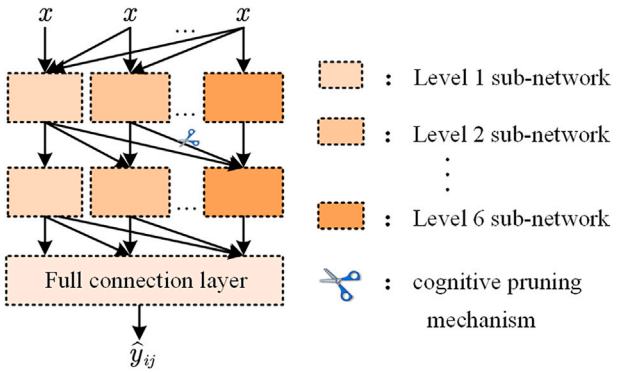


Fig. 4. The architecture of the progressive interaction network.

3.4.1. Exercise-cognitive level mapping matrix construction

According to Bloom's taxonomy [16], cognitive skills can be categorized into six levels: remembering, understanding, applying, analyzing, evaluating, and creating. Following the annotation approach proposed by MOOCRadar [14], we determine a series of representative verbs for each cognitive level. Based on these trigger words, each exercise is annotated to the highest cognitive level, and an exercise–cognitive level mapping matrix $C = \left(C_{jl} \in \{0,1\}\right)^{M\times 6}$ is constructed. This aims to conduct an in-depth study of how learners utilize different cognitive skills to comprehend and analyze exercises, as well as how they organize and apply various knowledge concepts to solve problems.

3.4.2. Diagnostic features fusion

The cognitive discrimination h^{disc} is extracted to participate in the simulation of subsequent interaction process, which can be denoted as:

$$h^{disc} = \sigma\left(\mathcal{F}\left(\text{Mask}(C) \circ \left(h^{prof_c} - h^{diff_c}\right)\right)\right) \tag{11}$$

where σ denotes the activation function, for which the Sigmoid function is specifically employed; $\mathcal{F}$ signifies an MLP; $\circ$ represents the operation of dot product; Mask$(\cdot)$ denotes the implementation of the cognitive mask mechanism, which is predicated on the principle that an exercise is designed to examine not only the explicitly labeled highest cognitive level but also all lower cognitive levels that align with the inherent progressive nature of human cognition.

For exercise j, the specific operation of the cognitive mask mechanism is expressed as:

$$\text{Mask}(C_{jl}): \quad C_{jt} = 1, \text{ for } t = 1, \ldots, l-1 \tag{12}$$

The diagnostic features are first fused to simulate the interaction between learners and exercises, which is formulated as:

$$x = Q \circ \left(h^{prof} - h^{diff}\right) \times h^{disc} \tag{13}$$

3.4.3. Progressive interaction network construction

For exercises corresponding to different cognitive levels, since they examine different cognitive skills, the cognitive processes of learners in solving these exercises are also different. The cognitive interaction process from low to high cognitive levels should be progressively complex. Drawing inspiration from the progressive neural networks [29,30], a progressive interaction network $\mathcal{F}_p$ is developed for interaction simulation, consisting of six distinct sub-networks, each corresponding to a cognitive level. Each sub-network is dedicated to learning a level-specific feature space, with the reuse of common interactive feature information from lower cognitive levels. In addition to receiving outputs from the previous layer of the current-level sub-network, it also shares the lower-level network parameters and aggregates outputs from all lower-level sub-networks.

The initial fused feature x is fed into different sub-networks for further interactions according to the cognitive levels presented by exercises. For exercise j, the selection of sub-networks is expressed as:

$$l_j = l, \quad \text{if } C_{jl} = 1 \tag{14}$$

where l_j represents the cognitive level index corresponding to the selected sub-network.

As shown in Fig. 4, the fundamental architecture of the progressive interaction network comprises two linear layers alongside a shared fully connected output layer, and follows the assumption of monotonicity [11]. Drawing insights from [31], a cognitive pruning mechanism is devised, which dynamically prunes non-essential neurons and connections during training process. The purpose is to streamline the network and precisely adapt to the demands of diagnostic tasks at varied cognitive levels.

Initially, all neurons across the network layers are unassigned and considered as empty neurons. Neurons are assigned to sub-networks based on the proportion of exercise responses at each cognitive level in the dataset. The pruning process can be denoted by

the following formulas:

$$\phi_m^d = \sum_{t=1}^{n_m} W_{mt} \tag{15}$$

$$P_m^{d\prime} = 0.99 P_m^d + 0.01 \cdot \left(Norm(\phi_m^d) - \rho \right) \tag{16}$$

For the neuron pruning, the importance of neuron m in the d-th layer is assessed by calculating the sum of its input weights, denoted as ϕ_m^d. W_{mt} represents the weight from neuron t in the previous layer to neuron m in the current layer, and n_m denotes the number of neurons in the previous layer connected to neuron m. $Norm$ denotes a linear normalization operation. P_m^d and $P_m^{d\prime}$ denote the pruning metrics from the previous and current epoch, respectively. ρ governs the pruning intensity.

Throughout the training process, the importance of each neuron is continuously tracked. If $P_m^d < 0$, the neuron m undergoes pruning. This pruning mechanism is implemented across all layers of the progressive interaction network, excluding the final layer. Through the progressive interaction network, the predicted performance $\hat{y}_{ij}$ of learner i on exercise j is obtained:

$$\hat{y}_{ij} = \sigma\left(\mathcal{F}_p(x) \right) \tag{17}$$

Ultimately, the cross-entropy loss function is employed to optimize the entire model through the minimization of the difference between the predicted performance $\hat{y}_{ij}$ and true response y_{ij}.

$$\text{Loss} = -\sum_{i=1}^{N} \sum_{j=1}^{J} \left[y_{ij} \log \hat{y}_{ij} + (1 - y_{ij}) \log \left(1 - \hat{y}_{ij} \right) \right] \tag{18}$$

By training with the above loss, all neural network parameters are optimized. h^{prof} and h^{prof_c} jointly serve as the assessment results of learners' cognitive states.

4. Experiments

4.1. Experimental purpose

Six experiments are designed to evaluate ICSCD: **Comparison experiments** are designed to evaluate ICSCD in normal and high unlearned concept ratio scenarios compared with eight classic and cutting-edge models.

Ablation experiments aim to verify the impact of different components on performance of ICSCD.

Training strategies experiments aim to verify the impact of different training strategies on the performance of ICSCD.

Visualization experiments are designed to analyze the relationship between learners' cognitive proficiency and their performance, as well as their ability to capture the associations between concepts.

Hyperparameter experiments are designed to explore the impact of neuron pruning intensity on performance.

Case study aims to analyze what personalized diagnostic information ICSCD can provide, and how to interpret the diagnostic results of the learner's cognitive states.

4.2. Experimental settings

4.2.1. Dataset descriptions

For the experimental evaluation of ICSCD, three representative educational datasets are selected: MOOCRadar [14], Junyi, and Eedi, which are widely adopted by advanced algorithms in the field. Table 2 displays detailed statistics and a brief introduction to the three datasets is as follows:

MOOCRadar (2023)[1]: MOOCRadar is a fine-grained, multi-aspect database developed based on MOOCCubeX [32]. It covers multidisciplinary fields such as engineering, law, medicine, and economics. It develops a suite of dedicated tools for fine-grained annotation of concepts and cognitive level labels of granularity.

Junyi (2020)[2]: The dataset Junyi comprises learners' exercise logs recorded between August 2018 and July 2019. These exercises are sourced from the mathematics test question repository of the Junyi Academy e-learning platform, with each exercise consisting of a single textual question.

Eedi (2020)[3]: The dataset Eedi encompasses learner response data related to mathematical problem-solving activities, which comes from the NeurIPS 2020 Education Challenge, built on four tasks. For the experimental analysis, the data from Tasks 3 and 4 are selected, which contain exercise text information.

All datasets include learner-exercise-concept mapping relations, as well as topic-concept mapping relations. Junyi offers annotated prerequisites by domain experts. The method outlined in [13] is employed to derive prerequisites for the concepts within the other two datasets. To achieve reliable and accurate diagnosis, learners are required to have sufficient response interaction records on target concepts or related concepts. Consistent with common practices in other methods, learners with fewer than 15 response records are

[1] https://github.com/THU-KEG/MOOC-Radar.

[2] https://www.kaggle.com/datasets/junyiacademy/learning-activity-public-dataset-by-junyi-academy/data.

[3] https://competitions.codalab.org/competitions/25449.

Table 2
The statistics of all datasets.

Dataset	MOOCRadar	Junyi	Eedi
#Learners	108	339	2202
#Exercises	729	1298	919
#Concepts	465	170	63
#Response logs	43,132	185,242	756,181
#Concepts per exercise	2.25	1	1
#Concepts per learner	265.87	57.04	39.50

Table 3
Details of experimental environment.

Environment	Description
GPU	NVIDIA GeForce RTX 4090, 24G
Python	v3.9.7
Pytorch	v1.12.1
Anaconda	V4.10.3

filtered out from the Eedi and Junyi datasets, while those with fewer than 200 response records are removed from the MOOCRadar dataset.

4.2.2. Experimental environment and conditions

Table 3 reports the details of the experimental environment. All experiments are implemented using PyTorch v1.12.1 and conducted on an NVIDIA GeForce RTX 4090 GPU with 24GB.

4.2.3. Implementation details

The network parameters of ICSCD are initialized utilizing Xavier initialization [33] and optimized through the utilization of the Adam optimizer [34]. The batch size is set to 256, and the learning rate is configured to 0.001. The number of HGAT layers is 2. P_i^d is initialized to 5. The exercise records are initially divided into six different groups based on the cognitive level of the exercise annotations. The data from each group are then split 80%/10%/10% for training/validation/testing. This data splitting method ensures the uniformity of cognitive levels in each batch of data.

4.2.4. Evaluation metrics

Unlike conventional supervised tasks, directly annotating labels for learners' true cognitive states is extremely difficult and costly, making the evaluation of cognitive diagnosis model performance challenging. Consequently, in line with most works, the cognitive diagnosis results are indirectly evaluated by predicting learners' performance. Specifically, four metrics recognized in the field are adopted. In terms of classification, *AUC* and *ACC* are used to evaluate prediction performance. *AUC* denotes the area enclosed by the coordinate axes under the ROC curve. The calculation of the metric *ACC* is defined as:

$$ACC = \frac{TP + TN}{TP + FP + FN + TN} \tag{19}$$

where *TP* (True Positive) represents the number of positive samples correctly classified, *TN* (True Negative) represents the number of negative samples correctly classified, *FP* (False Positive) represents the number of negative samples incorrectly classified, and *FN* (False Negative) represents the number of samples in which the true label is a positive example that is incorrectly judged as negative example.

From a regression standpoint, the root mean square error (*RMSE*) is used to measure the gap between the predicted performance $\hat{y}_{ij}$ and true response y_{ij}. The metric *RMSE* is defined as:

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(y_{ij} - \hat{y}_{ij}\right)^2} \tag{20}$$

where T represents the total number of samples.

The interpretability of the model hinges on the alignment between its diagnostic outcomes and the monotonicity assumption. In accordance with the monotonicity assumption, the probability of correct response to the exercise is monotonically increasing at any dimension of the learner's knowledge proficiency. Consistent with prior research [10], the degree of agreement (*DOA*) is employed to evaluate model interpretability. For the concept k, the DOA_k can be calculated as follows:

$$DOA_k = \frac{1}{Z_1}\sum_{a=1}^{N}\sum_{b=1}^{N}\mathbb{I}\left(h_{ak}^{prof} > h_{bk}^{prof}\right)\frac{\sum_{j=1}^{M}\mathbb{I}\left(Q_{jk}=1\right)\wedge J(j,a,b)\wedge\mathbb{I}\left(y_{aj} > y_{bj}\right)}{Z_0} \tag{21}$$

$$Z_0 = \sum_{j=1}^{M}\mathbb{I}\left(Q_{jk}=1\right)\wedge J(j,a,b)\wedge\mathbb{I}\left(y_{aj} \neq y_{bj}\right) \tag{22}$$

Table 4
Proportion of responses to exercises containing unlearned concepts in the test sets.

	MOOCRadar		Junyi		Eedi	
	NS	HUS	NS	HUS	NS	HUS
Response proportion	0.2870	0.7386	0.0010	0.8670	0.0234	0.9496

$$Z_1 = \sum_{a=1}^{N}\sum_{b=1}^{N} \mathbb{I}\left(h_{ak}^{prof} > h_{bk}^{prof}\right) \mathbb{I}\left(Z_0 > 0\right) \tag{23}$$

where $\mathbb{I}$(statement) is an indicator function that yields 1 if the statement is true or 0 otherwise. h_{ak}^{prof} signifies the proficiency of learner a on concept k. $J(j, a, b)$ indicates whether both a and b answered exercise j. DOA_k across all concepts are finally averaged, denoted as:

$$DOA = \frac{1}{K}\sum_{k=1}^{K} DOA_k \tag{24}$$

4.3. Comparison experiment

The dataset is divided using two strategies for constructing comparison scenarios: the Normal Scenario (NS) and High Unlearned concept ratio Scenario (HUS). In the normal scenario, the exercise records are randomly split following the data splitting method. In the high unlearned concept ratio scenario, the splitting algorithm proposed in [11] is applied to the datasets. This splitting algorithm maintains a fixed train/validation/test ratio, with priority given to assigning exercise records that contain unlearned concepts to the test set. Table 4 shows the proportion of unlearned concepts in the test set, where HUS splitting displays a significant increase compared to the proportion observed in NS.

Eight algorithms are reproduced as the comparison methods, which are detailed as follows:

MIRT (2009)[4] [17] is a multidimensional extension of IRT, which employs a probabilistic model to estimate the likelihood of a correct response based on a test taker's multidimensional ability.

DINA (2009)[5] [9], is a classic cognitive diagnosis model that characterizes each learner's knowledge proficiency using a binary vector and takes slips and guesses into account in the diagnostic process.

NCDM (2020)[6] [10] stands as one of the most widely adopted neural network-based cognitive diagnosis methods. It employs a multi-layer perceptron to capture intricate learner-exercise interactions and incorporates the monotonicity assumption to guarantee the interpretability of the model.

RCD (2021)[7] [13] focuses on the complex relationships among learners, exercises, and concepts, employing multi-layer attention networks to model prerequisite relations between concepts.

KaNCD (2022)[8] [11] utilizes a representation-based method to implicitly model knowledge associations, with the goal of mitigating inaccuracies in proficiency regarding unlearned concepts.

QCCDM (2023)[9] [15] is an advanced cognitive diagnosis model that integrates a structural causal model to capture the causality between learners' mastery on different concepts, enabling accurate inferences of their proficiency of rarely observed knowledge concepts.

ACD (2024)[10] [23] integrates learners' cognitive and affective states—such as frustration or confusion—into a unified framework, enhancing the accuracy and interpretability of knowledge proficiency estimation by modeling affective influences on guessing and slipping behaviors.

ORCDF (2024)[11] [24] constructs a novel response graph to encode answer interactions and designs a tailored response-aware graph convolution to capture discriminative response features, effectively alleviating mastery value over-smoothing while improving prediction accuracy and interpretability.

ICSCD and all the comparison methods are trained and tested in the same experimental environment using the official open-source code. Regarding the *DOA* metric, there is no comparison with MIRT and RCD due to the lack of clear correspondence between their latent features and concepts. In addition, the paired t-test results (p-v) of ICSCD and almost all baseline methods show that p-value< 0.05, which further proves that its performance improvement is statistically significant, and not due to random factors.

Table 5 reports the experimental results in the normal scenario. The key findings are as follows:

[4] https://github.com/bigdata-ustc/EduCDM/tree/main/EduCDM/MIRT.
[5] https://github.com/bigdata-ustc/EduCDM/tree/main/EduCDM/DINA.
[6] https://github.com/bigdata-ustc/Neural_Cognitive_Diagnosis-NeuralCD.
[7] https://github.com/bigdata-ustc/RCD.
[8] https://github.com/bigdata-ustc/Neural_Cognitive_Diagnosis-NeuralCD.
[9] https://github.com/ECNU-ILOG/QCCDM.
[10] https://github.com/zeng-zhen/ACD.
[11] https://github.com/lswhim/ORCDF.

Table 5
Experimental results of learner performance prediction on three datasets in the normal scenario.

Model	MOOCRadar(NS)					Junyi(NS)					Eedi(NS)				
	ACC	AUC	RMSE	DOA	*p-v*	ACC	AUC	RMSE	DOA	*p-v*	ACC	AUC	RMSE	DOA	*p-v*
MIRT	0.5690	0.6040	0.6323	–	0.0002	0.6524	0.6336	0.5317	–	0.0001	0.6514	0.7114	0.4831	–	0.0001
DINA	0.5675	0.7050	0.4889	0.8883	0.0001	0.7067	0.7043	0.4698	0.8156	0.0001	0.6422	0.7178	0.4961	0.8077	0.0001
NCDM	0.7782	0.8147	0.3942	0.8841	0.0184	0.7359	0.7461	0.4226	0.8231	0.0077	0.7023	0.7682	0.4421	0.8273	0.0238
RCD	0.7869	0.8248	0.3864	–	0.0266	0.7321	0.7436	0.4229	–	0.0386	0.7077	0.7767	0.4382	–	0.0244
KaNCD	0.7872	0.8261	0.3851	0.8968	0.0310	0.7325	0.7469	0.4232	0.8433	0.0400	0.7167	0.7853	0.4326	0.8618	0.0472
QCCDM	0.7896	0.8242	0.3851	0.8845	0.0242	0.7335	0.7452	0.4234	0.8340	0.0352	0.7177	0.7868	0.4321	0.8513	0.0399
ACD	0.7872	0.8259	0.3839	0.9064	0.0489	0.7313	0.7431	0.4234	0.8412	0.0511	0.7178	0.7856	0.4365	0.8596	0.0456
ORCDF	0.7795	0.8271	0.3820	0.8744	0.0433	0.7427	0.7507	0.4256	0.8502	0.0410	0.7159	0.7745	0.4361	0.8470	0.0312
ICSCD	**0.8041**	**0.8420**	**0.3744**	**0.9300**	-	**0.7413**	**0.7532**	**0.4178**	**0.8594**	-	**0.7235**	**0.7905**	**0.4290**	**0.8669**	-

Table 6
Experimental results of learner performance prediction on three datasets in high unlearned concept ratio scenario.

Model	MOOCRadar(HUS)					Junyi(HUS)					Eedi(HUS)				
	ACC	AUC	RMSE	DOA	*p-v*	ACC	AUC	RMSE	DOA	*p-v*	ACC	AUC	RMSE	DOA	*p-v*
MIRT	0.4792	0.5029	0.7045	–	0.0001	0.5660	0.5413	0.6190	–	0.0006	0.6381	0.6769	0.5150	–	0.0001
DINA	0.5080	0.6675	0.5270	0.8850	0.0008	0.5193	0.5683	0.5362	0.8121	0.0004	0.5068	0.5801	0.5772	0.7952	0.0001
NCDM	0.7703	0.7844	0.4075	0.8818	0.0216	0.7136	0.6521	0.4395	0.8199	0.0124	0.6044	0.6515	0.4826	0.8101	0.0237
RCD	0.7704	0.7875	0.4014	–	0.0170	0.7318	0.6847	0.4303	–	0.0288	0.6964	0.7594	0.4458	–	0.0371
KaNCD	0.7750	0.7882	0.4003	0.8906	0.0289	0.7218	0.6850	0.4326	0.8380	0.0430	0.6985	0.7617	0.4437	0.8613	0.0445
QCCDM	0.7689	0.7798	0.4096	0.8807	0.0325	0.7283	0.7077	0.4278	0.8444	0.0452	0.6993	0.7627	0.4433	0.8504	0.0523
ACD	0.7731	0.7868	0.4001	0.9070	0.0469	0.7307	0.7079	0.4272	0.8409	0.0473	0.6997	0.7600	0.4443	0.8552	0.0401
ORCDF	0.7679	0.7866	0.4052	0.9105	0.0484	0.7325	0.6950	0.4299	0.8511	0.0396	0.6906	0.7615	0.4440	0.8576	0.0389
ICSCD	**0.7800**	**0.7947**	**0.3960**	**0.9348**	-	**0.7371**	**0.7177**	**0.4236**	**0.8609**	-	**0.7051**	**0.7668**	**0.4408**	**0.8681**	-

(1) ICSCD outperforms the comparison methods across all metrics on the three datasets. This result can be mainly attributed to the fact that ICSCD simulates the progressive characteristics of cognitive process, enabling it to accurately capture learner interactive features at different levels of cognitive skills. Compared with RCD and QCCDM, which also model concept relations, ICSCD enhances the aggregation of relational information within the knowledge structure to obtain accurate concept representation, thereby facilitating the prediction performance.
(2) ICSCD exhibits a marked advancement in *DOA* metric, underscoring its superior diagnosis interpretability. This can be attributed to the fact that ICSCD can obtain more reliable knowledge proficiency, especially for unlearned concepts, thereby making the diagnosis results more consistent with the monotonicity assumption. In contrast, ICSCD shows a smaller advancement in *RMSE*, which can be attributed to the fact that the model exhibits relatively high sensitivity to extreme values or outliers. Specifically, the interaction data corresponding to exercises that involve unlearned concepts can be reasonably regarded as such outliers. ICSCD cannot completely eliminate the high prediction error caused by the lack of data, so the advancement is relatively small.
(3) ICSCD achieves the most substantial performance leap on the MOOCRadar dataset, while less advancement is made on the other two datasets. This is because ICSCD leverages both relation-level and node-level attention, which enables it to measure the importance of different concepts in exercises. In contrast to competing approaches including KaNCD, ACD and ORCDF, the proposed ICSCD framework exhibits superior capability in handling the complex one-to-many mapping between exercises and concepts within the MOOCRadar dataset, which further accounts for its outstanding overall performance.

Table 6 reports the experimental results in the high unlearned concept ratio scenario. Compared with the normal scenario, the performance drops significantly due to insufficient exercise records. ICSCD still outperforms the comparison methods across the board, especially compared to the two cutting-edge methods KaNCD and QCCDM designed specifically for the high unlearned concept ratio scenario. Specifically, although KaNCD attempts to capture implicit concept associations through representation-based techniques, it assumes that all concepts are interconnected, which may introduce noise and impair diagnostic outcomes. QCCDM relies on explicit prerequisite relations among concepts to assess learners' proficiency on unlearned concepts. However, when the paths between learned and unlearned concepts are long, relational information tends to degrade, reducing diagnostic accuracy. ICSCD constructs association paths among concepts under the same topic to enhance the relevance between learned and unlearned concepts, thereby reducing the loss of relational information, and enabling accurate assessment.

4.4. Ablation experiment

Three variants are designed to investigate the influence of different innovative components of ICSCD on performance, which are detailed as follows: **ICSCD w/o PIN**: It is obtained by replacing the progressive interaction network with a normal three-layer perceptron for interaction simulation.

ICSCD w/o CDE: It is a variant of ICSCD that removes the cognitive discrimination in the diagnostic feature extraction module.

Table 7
Ablation results of different ICSCD variants. The best results are marked in bold.

Model	MOOCRadar				Junyi				Eedi			
	ACC	AUC	RMSE	DOA	ACC	AUC	RMSE	DOA	ACC	AUC	RMSE	*DOA*
ICSCD	**0.8041**	**0.8420**	**0.3744**	**0.9300**	**0.7413**	**0.7532**	**0.4178**	**0.8609**	**0.7235**	**0.7905**	**0.4290**	**0.8681**
-w/o PIN	0.7987	0.8407	0.3755	0.9284	0.7402	0.7523	0.4183	0.8576	0.7191	0.7880	0.4305	0.8662
-w/o CDE	0.7976	0.8351	0.3783	0.9283	0.7400	0.7523	0.4182	0.8580	0.7217	0.7877	0.4300	0.8652
-w/o CLR	0.7968	0.8371	0.3773	0.9284	0.7405	0.7529	0.4181	0.8575	0.7209	0.7882	0.4305	0.8634
-w/o PC	0.8014	0.8413	0.3749	0.9291	0.7408	0.7528	0.4180	0.8591	0.7223	0.7900	0.4295	0.8643
-w/o *Pr*	0.8001	0.8411	0.3752	0.9299	0.7405	0.7529	0.4181	0.8601	0.7222	0.7900	0.4296	0.8669
-w. Drop	0.8006	0.8411	0.3754	0.9289	0.7404	0.7525	0.4181	0.8601	0.7221	0.7902	0.4297	0.8650

Table 8
Experimental results of different training strategies. The best results are marked in bold.

Model	MOOCRadar				Junyi				Eedi			
	ACC	AUC	RMSE	DOA	ACC	AUC	RMSE	DOA	ACC	AUC	RMSE	*DOA*
ICSCD w. s1	0.8003	0.8412	0.3753	0.9300	0.7398	0.7521	0.4183	0.8584	0.7192	0.7878	0.4309	0.8678
ICSCD w. s2	0.7989	0.8364	0.3778	0.9295	0.7380	0.7486	0.4197	0.8607	0.7184	0.7860	0.4320	0.8664
ICSCD w. s3	**0.8041**	**0.8420**	**0.3744**	**0.9300**	**0.7413**	**0.7532**	**0.4178**	**0.8609**	**0.7235**	**0.7905**	**0.4290**	**0.8681**

ICSCD w/o CLR: It is a variant of ICSCD achieved by removing the topic–concept bipartite graph within the node embedding representation module.

In addition, three variants are designed to deeply explore the impact of different components in the progressive interaction network on performance, as follows: **ICSCD w/o PC**: It is obtained by removing the progressive connection between different sub-networks within the progressive interaction network. That is, each cognitive level sub-network is an independent small multi-layer perceptron.

ICSCD w/o Pr: It is obtained by not applying the cognitive pruning mechanism in the progressive interaction network.

ICSCD w. Drop: It is obtained by replacing the cognitive pruning mechanism with random dropout. The dropout rate is set to 0.5 in the experiment.

Table 7 details the experimental results and yields the following findings.

(1) Removing any of the progressive interaction network (-w/o PIN), the cognitive discrimination (-w/o CDE), and topic–concept bipartite graph (-w/o CLR) results in a decreased performance. Notably, the exclusion of the cognitive discrimination extraction module yields the most significant performance degradation. Neglecting it causes the model to fail to identify learners' cognitive skill differences when solving exercises at different cognitive levels, resulting in a bias in the fitting of the interaction process.
(2) Using a single three-layer perceptron as the interaction function (-w/o PIN) results in decreased performance, since single three-layer perceptron fails to distinguish interaction patterns at different cognitive levels. Replacing the progressive connection with independent sub-network structures (-w/o PC) to model the cognitive process at each level independently overlooks the sharing of information between different cognitive levels. The progressive connection ensures the reliable transmission and sharing of lower-level interaction features while learning unique higher-level interaction features. This is conducive to the reuse of interactive features across various cognitive skills, so that the feature information learned in low cognitive level tasks to support for higher-level tasks, leading provides an overall elevation.
(3) Compared to applying the cognitive pruning mechanism based on neuron importance, using random dropout (-w. Drop) or absent pruning (-w/o Pr) within the progressive interaction network leads to performance degradation. This is because the absence of the pruning can result in neuron redundancy, negatively impacting task accuracy, while indiscriminate random pruning may harm critical neurons. The neuron importance-based approach selectively retains critical neurons essential for diagnosis while discarding redundant neurons contributes to streamlining the network and bolstering the diagnostic performance.

4.5. Training strategies experiments

Three different training strategies are designed to delve into the impact of training strategies on model performance. The first is to train in a progressive manner, i.e., train with data containing exercises at the first cognitive level and then increase the data level by level. The second strategy is random training, where batches of data are randomly selected. The third strategy involves a combination of the first two strategies: initially training in a progressive manner, followed by random training. The variants corresponding to the three training strategies are represented as ICSCD w.s1, ICSCD w.s2, and ICSCD w.s3.

The experimental results are shown in Table 8. Utilizing only random or entirely progressive training strategies leads to suboptimal outcomes, while the fusion of two strategies produces the best model performance. This is attributed to the fact that the initial progressive training facilitates the model's learning of the data distribution and underlying patterns at a single cognitive level, supplemented by subsequent random training to effectively broaden the model's generalization capacity. Single random training methods may cause the model to converge to a local optimal solution, while entirely progressive training methods may fall into overfitting on a single cognitive level task. The proposed model is trained using the third strategy by default.

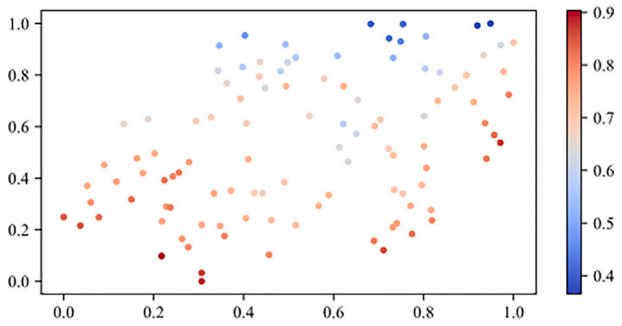


Fig. 5. t-SNE visualization of cognitive proficiency colored with average score on MOOCRadar.

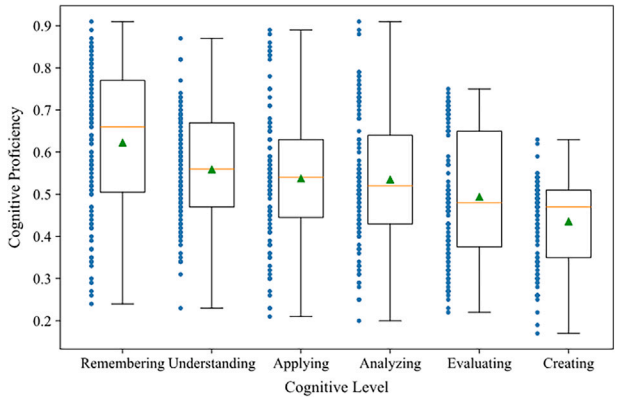


Fig. 6. The distribution of learners' proficiency in different cognitive skills on MOOCRadar.

4.6. Visualization experiments

4.6.1. Cognitive proficiency

T-SNE [35] is employed to visualize cognitive proficiency vectors to gain an analysis of the relationship between learners' abilities in different cognitive skills and their academic performance. The average answer score of each corresponding learner is utilized to color-code the vectors on MOOCRadar dataset.

As depicted in Fig. 5, it is clearly observed that the distribution of learners' cognitive proficiency is closely related to their performance, indicating that differences in learners' cognitive skills when solving problems significantly affect their academic performance. Samples located further toward the lower-left quadrant correspond to learners with higher cognitive proficiency, who typically achieve better academic performance. Higher cognitive proficiency reflects learners' strengths in understanding, analyzing, etc., and they tend to be capable of tackling diverse learning tasks and achieving outstanding accomplishments.

Furthermore, an experimental analysis is conducted on the ability distribution of learners at different cognitive skills. Fig. 6 illustrates the cognitive proficiency of 108 learners on the MOOCRadar dataset across six cognitive skills. The results display a gradual decrease in the average cognitive proficiency from the lowest to the highest level of cognitive skills. This trend further verifies the progressive nature of cognitive skills from simple to complex, and suggests that learners may face greater difficulty in tackling increasingly intricate cognitive tasks.

4.6.2. Concept relations

The Euclidean distance D of the embeddings of concept pairs with different relations is computed to measure their correlations, including direct prerequisite (1-hop prerequisite), 2-hop prerequisite, 3-hop prerequisite, topic relation, and no relation. Concerning the topic relations between concepts, two concepts that belong to the same topic are paired. In cases where a pair of concepts does not have the above four relations, they are considered to lack an explicit relation.

Fig. 7 represents the experimental results conducted on the MOOCRadar dataset. Observations from the figure reveal that concepts with prerequisites as well as topic relations exhibit smaller distances compared to concepts without relations. This suggests that ICSCD captures the concept relations. Concepts with direct prerequisites show the smallest distances, as understanding a prerequisite concept is essential for grasping subsequent related concepts, showing a strict sequential relatedness. The distance increases with the increase in the number of hops, verifying that the association decays as the inference path grows. Topic relations show a smaller distance than the multi-hop prerequisites, verifying the effectiveness of our method in inferring the proficiency of unlearned concepts.

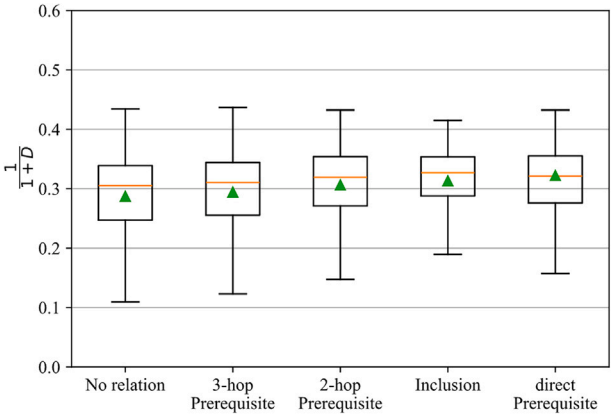


Fig. 7. Results of inter-concept correlations on MOOCRadar.

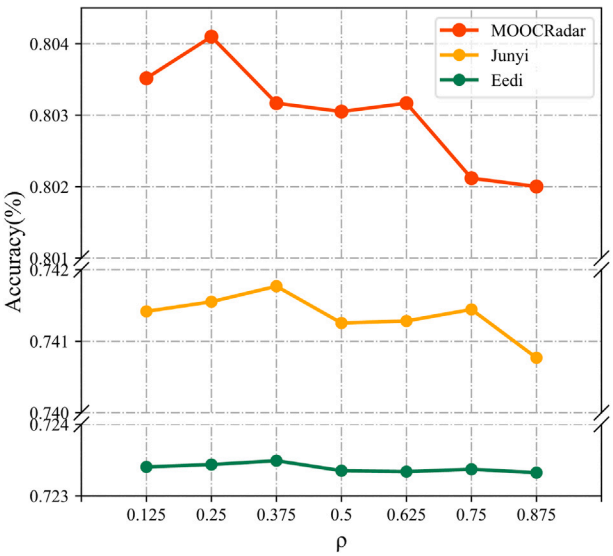


Fig. 8. The influence of the parameter ρ on model performance.

4.7. Parameter experiment

Hyperparameter experiments are conducted to investigate the influence of the critical parameter ρ on model performance.

The hyperparameter ρ plays a pivotal role in controlling the extent of neuron pruning within progressive neural networks, directly influencing the network's performance by determining which neurons are retained and which are eliminated. Fig. 8 illustrates the variation of *ACC* with respect to ρ across three datasets. Experimental results show that moderate pruning is advantageous, as eliminating redundant neurons renders the network specific and precise for the given tasks. As ρ increases beyond a certain threshold, the pruning becomes excessively aggressive, leading to the elimination of crucial neurons that are vital for capturing the intricate patterns. This over-pruning results in a decline in performance. The negative impact of excessive pruning was particularly pronounced in complex datasets (such as MOOCRadar, which encompass a broader spectrum of concepts and where each exercise contains multiple concepts). The intricate interactive dynamics within this dataset necessitate a greater number of neurons for accurate modeling, rendering excessive pruning particularly detrimental.

Additionally, the pruning rate of neurons is investigated (experimented on the MOOCRadar dataset), and the results are shown in Fig. 9. Throughout the initial phases of training, redundant neurons within the network are gradually removed, leading to a rise in the proportion of pruned neurons. With successive training iterations, the remaining neurons gradually converge towards optimal configurations, so the proportion of pruned neurons tends to stabilize. This observed trend aligns with the rational progression of learning process demonstrated by neuroscience research [36], wherein neurons and synapses are moderately pruned based on external stimuli, ultimately emerging into a delicate and sparse network.

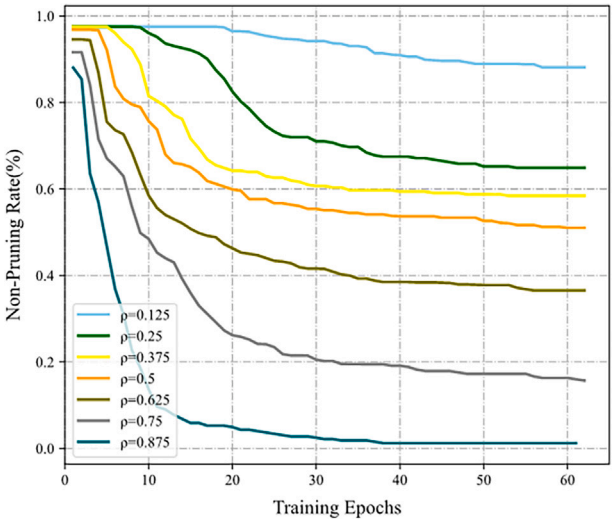
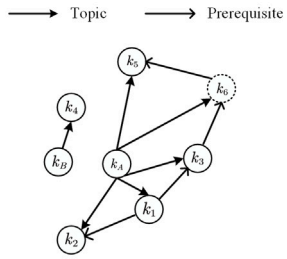


Fig. 9. The rate of neurons that are not pruned under different ρ on MOOCRadar during the training process.

Exercise	Cognitive Level	Knowledge Concept					Response	Prediction	
		k_1	k_2	k_3	k_4	k_5		QCCDM	ICSCD
e_1	L_2	1	0	0	1	0	✓ | ✓	✓ | ✓	✓ | ✓
e_2	L_1	0	0	1	0	0	✓ | ✗	✓ | ✗	✓ | ✗
e_3	L_2	0	0	0	0	1	✗ | ✓	✗ | ✓	✗ | ✓
e_4	L_3	0	1	0	0	0	✓ | ✓	✓ | ✓	✓ | ✓
e_5	L_5	1	1	0	0	0	✗ | ✗	✓ | ✗	✗ | ✗

(a)



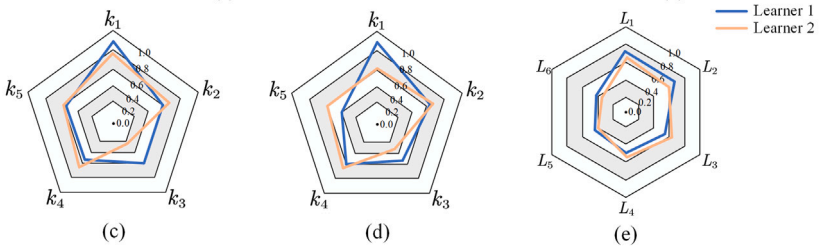


Fig. 10. An example of two learners' diagnostic reports. (a) The levels of cognitive skills and concepts examined by five exercises completed by Learners 1 and 2, as well as their actual responses and responses predicted by different models; (b) the relations between these concepts, where k_A and k_B represent the topics to which they belong; k_3, k_5 and k_6 are all unlearned concepts. Especially for k_6, the two learners did not answer the related exercises (there are no interaction logs in either the training set or the test set); (c) knowledge proficiency of QCCDM diagnostic outputs; (d) knowledge proficiency of ICSCD diagnostic outputs; (e) cognitive proficiency of ICSCD diagnostic outputs.

4.8. Case study

The application of cognitive diagnosis in intelligent education aims to furnish learners with comprehensive diagnostic reports. These reports serve as invaluable tools for uncovering individualized learning challenges and inspiring learners to undertake tailored adjustments. To this end, two learners are randomly selected from the MOOCRadar dataset to generate their diagnostic reports using QCCDM and ICSCD, and the diagnostic results are shown in Fig. 10.

From the perspective of interpretability, when a learner has a higher proficiency in a concept than another learner, he is more likely to answer the exercises related to it correctly. For e_3, Learner 2 provides a correct response, whereas Learner 1 makes an error. However, according to the diagnostic outcomes of QCCDM, the proficiency of both learners concerning k_5 in e_3 appears comparable, rendering an inadequate explanation for this result. This misjudgment may stem from the reliability of proficiency inference for unlearned concept k_5. QCCDM relies on a long inference path ($k_1 \rightarrow k_3 \rightarrow k_6 \rightarrow k_5$), where relational information progressively weakens, resulting in unreliable outcomes. ICSCD establishes a direct relational constraint path ($k_A \rightarrow k_5$), which concludes that Learner 2 has a high proficiency in k_5 and hence answers correctly. In contrast, Learner 1 has a lower proficiency, resulting in an incorrect answer.

It is noteworthy that ICSCD also assesses learners' proficiency at different cognitive skills, identifying the strengths and weaknesses of the individual's cognitive skills. In the case of e_5, QCCDM yields a positive prediction for Learner 1 that contradicted their actual response. This predictive discrepancy may stem from QCCDM's primary reliance on Learner 1's demonstrated high proficiency in k_1 and k_2 during the diagnostic process, while neglecting other potential cognitive factors. Instead, ICSCD considers the examination of high level cognitive skills inherent in e_5. Although Learner 1 performs well on the basic concepts (k_1 and k_2), ICSCD recognizes their weak proficiency at the L_5 level (evaluating skill), indicating that the learner lacks the ability to examine the text information in the exercise and make judgments, arriving at a negative prediction that aligned with the actual response.

5. Discussions

A novel cognitive diagnosis model is presented in this study, which optimizes the effect of cognitive diagnosis by identifying different cognitive skills examined by exercises and refining the interaction process simulation between learners and exercises. Specifically, we construct a mapping between exercises and cognitive levels, enabling a refined subdivision of the cognitive skills examined by each exercise, and extract the corresponding feature information to construct a progressive interaction network adapted to different cognitive levels for interaction simulation. Furthermore, we leverage topic relations to enrich concept embedding representations for reliable proficiency inference. Extensive experiments on real-world datasets demonstrate that ICSCD not only exhibits high effectiveness but also broadens the diagnostic framework by introducing an assessment focused on cognitive skills, providing a more comprehensive and detailed cognitive analysis of learners' performance.

ICSCD is pioneering in modeling and validating cognitive skills as a critical factor influencing learning performance, underscoring the necessity of incorporating these skills into cognitive diagnostic frameworks. It divides the cognitive process of learners' interaction with exercises into six progressive levels, delving into the nuanced differences in deeper cognitive operations involved rather than merely focusing on the categories of concepts acquired by learners. Additionally, the model aligns with extensive research focused on concept proficiency assessment, enabling more precise identification of learners' specific areas of weakness. Unlike other models, ICSCD also supports effective assessment of topic proficiency. This feature enables us to observe learners' overall mastery of related concept clusters or knowledge modules from a macro perspective, thereby offering a broader view for comprehensively understanding learners' overall learning progress. In practical applications, ICSCD allows for tracking learners' cognitive states at both the cognitive skills level and across varying knowledge granularities, laying a robust theoretical foundation and offering practical guidance for more nuanced and personalized instruction.

ICSCD is not intended to replace human expert diagnosis; rather, it serves as a powerful auxiliary tool within intelligent educational systems. In practical large-scale educational applications, such as massive online courses and long-term learning tracking scenarios, this model offers efficient and rapid proficiency diagnostic support. It helps alleviate the burden on educators, provides immediate feedback to learners, and facilitates large-scale adaptive interventions, thereby demonstrating significant practical value.

6. Conclusion and future work

This paper develops a novel cognitive diagnosis model, ICSCD, which for the first time unifies the proficiency assessment of concepts and cognitive skills into a diagnostic framework. An exercise-cognitive level mapping is devised to distinguish the cognitive skills examined by exercises. A progressive interaction network is designed coupled with a cognitive pruning mechanism to simulate the interaction process of different cognitive levels. Furthermore, a concept node embedding representation method that incorporates topic associations is developed to infer reliable knowledge proficiency, especially for unlearned concepts. Extensive experiments conducted on three real-world datasets demonstrate the superiority of ICSCD. It furnishes ability assessment reports that encompass diverse cognitive levels, offering an insight into strengths and defects of the individual's cognitive skills for tailored guidance.

However, ICSCD exhibits limitations in its assumption of learners maintaining a constant cognitive status in static testing scenarios, disregarding the dynamic nature of cognitive states throughout the learning journey. In light of this limitation, the varying characteristics of learners' cognitive states intend to be mined to achieve dynamic cognitive diagnosis. Additionally, future personalized learning recommendations ought to encompass not merely the identification of learners' weaknesses in knowledge proficiency but also the intricate tailoring to their current ability differences at various cognitive skills. To this end, the future work endeavors to delve into nuanced personalized strategies for recommending learning materials and pathways, with the goal of meticulously adapting to each learner's unique needs, fostering their overall learning efficiency.

CRediT authorship contribution statement

Xiaonan Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology. **Senlin Luo:** Supervision, Resources, Funding acquisition, Formal analysis. **Jinjie Zhou:** Validation, Software, Data curation. **Deshan Yang:** Writing – review & editing, Validation, Software. **Limin Pan:** Supervision, Methodology, Conceptualization.

Funding

This research is partially supported by the 242 National Information Security Projects, PR China under Grant 2020A065.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] H. Burns, C.A. Luckhardt, J.W. Parlett, C.L. Redfield, Intelligent Tutoring Systems: Evolutions in Design, Psychology Press, 2014.
[2] H.A. Alrakhawi, N. Jamiat, S.S. Abu-Naser, Intelligent tutoring systems in education: a systematic review of usage, tools, effects and evaluation, J. Theor. Appl. Inf. Technol. 101 (2023) 1205–1226.
[3] S. Yang, H. Wei, H. Ma, Y. Tian, X. Zhang, Y. Cao, Y. Jin, Cognitive diagnosis-based personalized exercise group assembly via a multi-objective evolutionary algorithm, IEEE Trans. Emerg. Top. Comput. Intell. 7 (2023) 829–844.
[4] G. Zhang, G. Yuan, Z. Xu, Y. Zhang, J. Ren, Z. Deng, D. Cheng, Harnessing LLM for noise-robust cognitive diagnosis in web-based intelligent education systems, in: Proceedings of the ACM Web Conference 2026, 2026, pp. 4093–4104.
[5] Y. Liu, T. Zhang, X. Wang, G. Yu, T. Li, New development of cognitive diagnosis models, Front. Comput. Sci. 17 (2023) 171604.
[6] F. Liu, X. Hu, S. Liu, C. Bu, L. Wu, Meta multi-agent exercise recommendation: a game application perspective, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023, pp. 1441–1452.
[7] C. Wang, H. Zhu, C. Zhu, X. Zhang, E. Chen, H. Xiong, Personalized employee training course recommendation with career development awareness, in: Proceedings of the Web Conference 2020, 2020, pp. 1648–1659.
[8] S.E. Embretson, S.P. Reise, Item Response Theory for Psychologists, Psychology Press, 2013.
[9] J. De La Torre, DINA model and parameter estimation: a didactic, J. Educ. Behav. Stat. 34 (2009) 115–130.
[10] F. Wang, Q. Liu, E. Chen, Z. Huang, Y. Chen, Y. Yin, Z. Huang, S. Wang, Neural cognitive diagnosis for intelligent education systems, in: Proceedings of the AAAI Conference on Artificial Intelligence, 34, 2020, pp. 6153–6161.
[11] F. Wang, Q. Liu, E. Chen, Z. Huang, Y. Yin, S. Wang, Y. Su, NeuralCD: a general framework for cognitive diagnosis, IEEE Trans. Knowl. Data Eng. 35 (2022) 8312–8327.
[12] Y. Su, Z. Cheng, J. Wu, Y. Dong, Z. Huang, L. Wu, E. Chen, S. Wang, F. Xie, Graph-based cognitive diagnosis for intelligent tutoring systems, Knowl.-Based Syst. 253 (2022) 109547.
[13] W. Gao, Q. Liu, Z. Huang, Y. Yin, H. Bi, M.-C. Wang, J. Ma, S. Wang, Y. Su, RCD: relation map driven cognitive diagnosis for intelligent education systems, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2021, pp. 501–510.
[14] J. Yu, M. Lu, Q. Zhong, Z. Yao, S. Tu, Z. Liao, X. Li, M. Li, L. Hou, H.-T. Zheng, et al., MoocRadar: a fine-grained and multi-aspect knowledge repository for improving cognitive student modeling in MOOCs, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2023, pp. 2924–2934.
[15] S. Liu, H. Qian, M. Li, A. Zhou, QCCDM: a q-augmented causal cognitive diagnosis model for student learning, in: ECAI 2023, IOS Press, 2023, pp. 1536–1543.
[16] D.R. Krathwohl, A revision of bloom's taxonomy: an overview, Theory pract. 41 (2002) 212–218.
[17] T.A. Ackerman, Multidimensional item response theory models, in: Wiley StatsRef: Statistics Reference Online, 2014, pp. 1–14.
[18] P. Zhan, H. Jiao, D. Liao, Cognitive diagnosis modelling incorporating item response times, Br. J. Math. Stat. Psychol. 71 (2018) 262–286.
[19] S. Pouyanfar, S. Sadiq, Y. Yan, H. Tian, Y. Tao, M.P. Reyes, M.-L. Shyu, S.-C. Chen, S.S. Iyengar, A survey on deep learning: algorithms, techniques, and applications, ACM comput. surv. (CSUR) 51 (2018) 1–36.
[20] H. Yang, T. Qi, J. Li, L. Guo, M. Ren, L. Zhang, X. Wang, A novel quantitative relationship neural network for explainable cognitive diagnosis model, Knowl.-Based Syst. 250 (2022) 109156.
[21] L. Song, M. He, X. Shang, C. Yang, J. Liu, M. Yu, Y. Lu, A deep cross-modal neural cognitive diagnosis framework for modeling student performance, Expert Syst. Appl. 230 (2023) 120675.
[22] H. Ma, C. Wang, H. Zhu, S. Yang, X. Zhang, X. Zhang, Enhancing cognitive diagnosis using un-interacted exercises: a collaboration-aware mixed sampling approach, in: Proceedings of the AAAI Conference on Artificial Intelligence, 38, 2024, pp. 8877–8885.
[23] S. Wang, Z. Zeng, X. Yang, K. Xu, X. Zhang, Boosting neural cognitive diagnosis with student's affective state modeling, in: Proceedings of the AAAI Conference on Artificial Intelligence, 38, 2024, pp. 620–627.
[24] H. Qian, S. Liu, M. Li, B. Li, Z. Liu, A. Zhou, ORCDF: an oversmoothing-resistant cognitive diagnosis framework for student learning in online education systems, in: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024, pp. 2455–2466.
[25] X. Li, S. Guo, J. Wu, C. Zheng, An interpretable polytomous cognitive diagnosis framework for predicting examinee performance, Inf. Process. Manag. 62 (2025) 103913.
[26] X. Wang, C. Huang, J. Cai, L. Chen, Using knowledge concept aggregation towards accurate cognitive diagnosis, in: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, 2021, pp. 2010–2019.
[27] Y. Sheng, J. Tang, W. Zeng, X. Zhao, Y. Tan, Enhancing cognitive diagnosis via LLM-driven heterogeneous concept graph construction, Engineering (2026) 1–17.
[28] T. Yang, L. Hu, C. Shi, H. Ji, X. Li, L. Nie, HGAT: heterogeneous graph attention networks for semi-supervised short text classification, ACM Trans. Inf. Syst. (TOIS) 39 (2021) 1–29.
[29] A.A. Rusu, N.C. Rabinowitz, G. Desjardins, H. Soyer, J. Kirkpatrick, K. Kavukcuoglu, R. Pascanu, R. Hadsell, Progressive neural networks, arXiv preprint arXiv:1606.04671, 2016.
[30] H.M. Fayek, L. Cavedon, H.R. Wu, Progressive learning: a deep learning framework for continual learning, Neural Netw. 128 (2020) 345–357.
[31] B. Han, F. Zhao, Y. Zeng, W. Pan, G. Shen, Enhancing efficient continual learning with dynamic structure development of spiking neural networks, arXiv preprint arXiv:2308.04749, 2023.
[32] J. Yu, Y. Wang, Q. Zhong, G. Luo, Y. Mao, K. Sun, W. Feng, W. Xu, S. Cao, K. Zeng, et al., MOOCCubeX: a large knowledge-centered repository for adaptive learning in MOOCs, in: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, 2021, pp. 4643–4652.
[33] X. Glorot, Y. Bengio, Understanding the difficulty of training deep feedforward neural networks, in: Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics, JMLR Workshop and Conference Proceedings, 2010, pp. 249–256.
[34] D.P. Kingma, J. Ba, Adam: a method for stochastic optimization, arXiv preprint arXiv:1412.6980, 2014.
[35] L. Van der Maaten, G. Hinton, Visualizing data using t-SNE, J. Mach. Learn. Res. 9 (2008).
[36] F. Zhao, Y. Zeng, J. Bai, Toward a brain-inspired developmental neural network based on dendritic spine dynamics, Neural Comput. 34 (2022) 172–189.